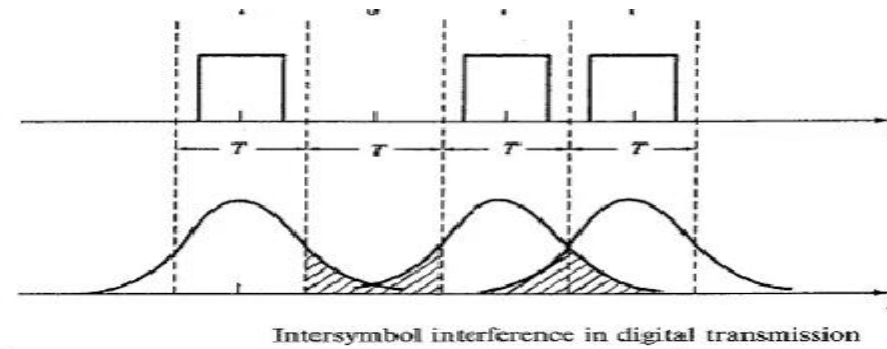


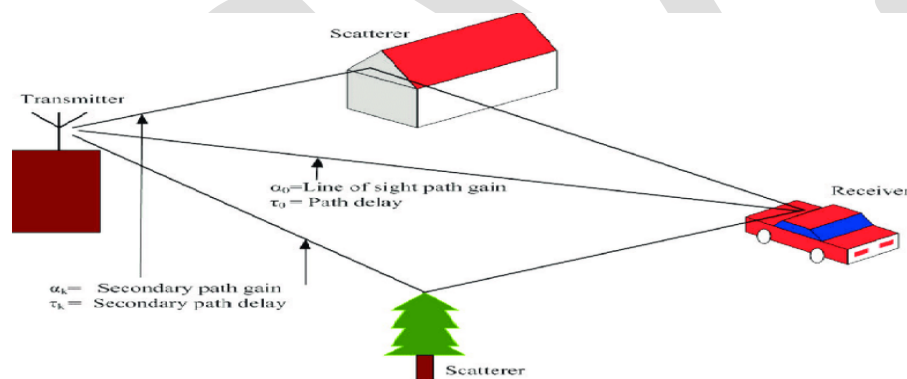
MODULE 5

INTER SYMBOL INTERFERENCE

Inter symbol interference (ISI) is when signals overlap and mix together, causing confusion in the information being sent. By managing the timing and using special techniques like spacing filtering and equalization (smoothing the wave) ISI can be reduced.



ISI is usually caused by multipath propagation or the inherent linear or non-linear frequency response of a communication channel causing successive symbols to blur together.



The source of bit errors in a base band-pulse transmission is inter symbol interference (ISI), which arises when the communication channel is dispersive. When we say a channel is dispersive, we mean the channel has a frequency dependent amplitude spectrum.

The simplest example of a dispersive channel is the band-limited channel. For example, a brick wall band-limited channel passes all frequencies $|f| < W$ without distortion, while it blocks all frequencies $|f| > W$.

For the baseband transmission of digital data, the use of discrete pulse-amplitude modulation (PAM) is the most efficient one in terms of power and bandwidth utilization.

In PAM, the amplitude of each pulse represents the data being transmitted. This makes it easy to visualize how different pulses can interfere with each other.

The pulses in PAM are distinct and their shapes can be easily observed, which helps in illustrating how they overlap and cause ISI.

Consider then a *baseband binary PAM system*, a generic form of which is shown in Figure 8.8. The incoming binary sequence $\{b_k\}$ consists of symbols 1 and 0, each of duration T_b . The *pulse-amplitude modulator* transforms this binary sequence into a new sequence of short pulses (approximating a unit impulse), whose amplitude a_k is represented in the polar form

$$a_k = \begin{cases} +1 & \text{if symbol } b_k \text{ is 1} \\ -1 & \text{if symbol } b_k \text{ is 0} \end{cases} \quad (8.42)$$

The sequence of short pulses so produced is applied to a *transmit filter* of impulse response $g(t)$, producing the transmitted signal (see Section 2.11).

$$s(t) = \sum_k a_k g(t - kT_b) \quad (8.43)$$

The signal $s(t)$ is modified as a result of transmission through the *channel* of impulse response $h(t)$. In addition, the channel adds random noise to the signal at the receiver input. The noisy signal $x(t)$ is then passed through a *receive filter* of impulse response $c(t)$. The resulting filter output $y(t)$ is sampled *synchronously* with the transmitter, with the sampling instants being determined by a *clock* or *timing signal* that is usually extracted from the receive-filter output. Finally, the sequence of samples thus obtained is used to reconstruct the original data sequence by means of a *decision device*. Specifically, the amplitude of each sample is compared to a *threshold* λ . If the threshold λ is exceeded, a decision is made in favor of symbol 1. If the threshold λ is not exceeded, a decision is made in favor of symbol 0. If the sample amplitude equals the threshold exactly, the flip of a fair coin will determine which symbol was transmitted (i.e., the receiver simply makes a guess).

The receive filter output is written as

$$y(t) = \mu \sum_k a_k p(t - kT_b) + n(t) \quad (8.44)$$

where μ is a scaling factor, and the pulse $p(t)$ is to be defined. To be precise, an arbitrary time delay t_0 should be included in the argument of the pulse $p(t - kT_b)$ in Eq. (8.44) to represent the effect of transmission delay through the system. To simplify the exposition, we have put this delay equal to zero in Eq. (8.44) without loss of generality.

The scaled pulse $\mu p(t)$ is obtained by a double convolution involving the impulse response $g(t)$ of the transmit filter, the impulse response $h(t)$ of the channel, and the impulse response $c(t)$ of the receive filter, as shown by

$$\mu p(t) = g(t) \star h(t) \star c(t) \quad (8.45)$$

where the star denotes convolution. We assume that the pulse $p(t)$ is *normalized* by setting

$$p(0) = 1 \quad (8.46)$$

which justifies the use of μ as a scaling factor to account for amplitude changes incurred in the course of signal transmission through the system.

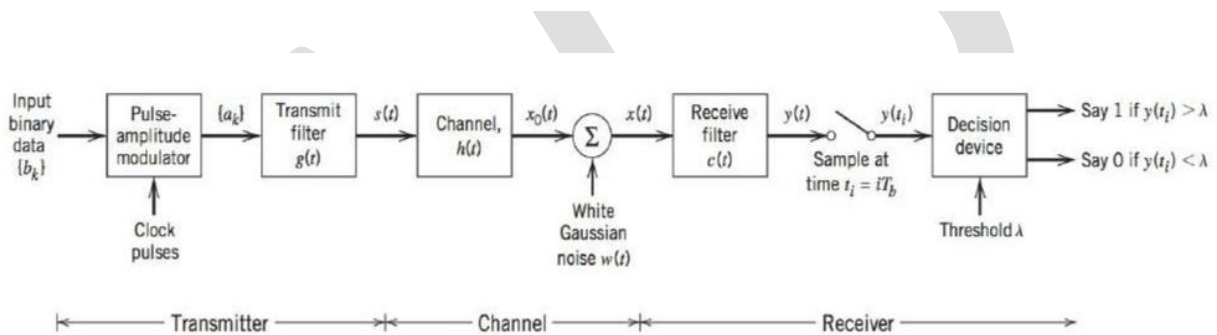


FIGURE 8.8 Baseband binary data transmission system.

Since the convolution in the time domain is transformed into multiplication in the frequency domain, we may use the Fourier transform to change Eq. (8.45) into the equivalent form

$$\mu P(f) = G(f)H(f)C(f) \quad (8.47)$$

where $P(f)$, $G(f)$, $H(f)$, and $C(f)$ are the Fourier transforms of $p(t)$, $g(t)$, $h(t)$, and $c(t)$, respectively.

Finally, the term $n(t)$ in Eq. (8.44) is the noise produced at the output of the receive filter due to the additive noise $w(t)$ at the receiver input. It is customary to model $w(t)$ as a white Gaussian noise of zero mean.

The receive filter output $y(t)$ is sampled at time $t_i = iT_b$ (with i taking on integer values), yielding [in light of Eq. (8.46)]

$$\begin{aligned} y(t_i) &= \mu \sum_{k=-\infty}^{\infty} a_k p[(i-k)T_b] + n(t_i) \\ &= \mu a_i + \mu \sum_{\substack{k=-\infty \\ k \neq i}}^{\infty} a_k p[(i-k)T_b] + n(t_i) \end{aligned} \quad (8.48)$$

In Eq. (8.48), the first term μa_i represents the contribution of the i th transmitted bit. The second term represents the residual effect of all other transmitted bits on the decoding of the i th bit; this residual effect due to the occurrence of pulses before and after the sampling instant t_i is called intersymbol interference (ISI). The last term $n(t_i)$ represents the noise sample at time t_i .

In the absence of both ISI and noise, we observe from Eq. (8.48) that

$$y(t_i) = \mu a_i$$

which shows that, under these ideal conditions, the i th transmitted bit is decoded correctly. The unavoidable presence of ISI and noise in the system, however, introduces errors in the decision device at the receiver output. Therefore, in the design of the transmit and receive filters, the objective is to minimize the effects of noise and ISI and thereby deliver the digital data to its destination with the smallest error rate possible.

Eye pattern

The eye diagram is a crucial tool in digital communication for evaluating the performance and quality of transmitted signals.

The eye diagram is a powerful and essential tool in digital communication for visualizing and analyzing signal quality. It helps in detecting and diagnosing issues, optimizing performance, and ensuring the reliability and integrity of communication systems. By providing a clear picture of how well the signals are transmitted and received, the eye diagram plays a critical role in maintaining efficient and effective digital communication.

Here's why it's significant:

Clarity of Signal: The eye diagram visually shows how well the signal is being transmitted. An open and clear "eye" pattern indicates that the signal is strong and easily distinguishable.

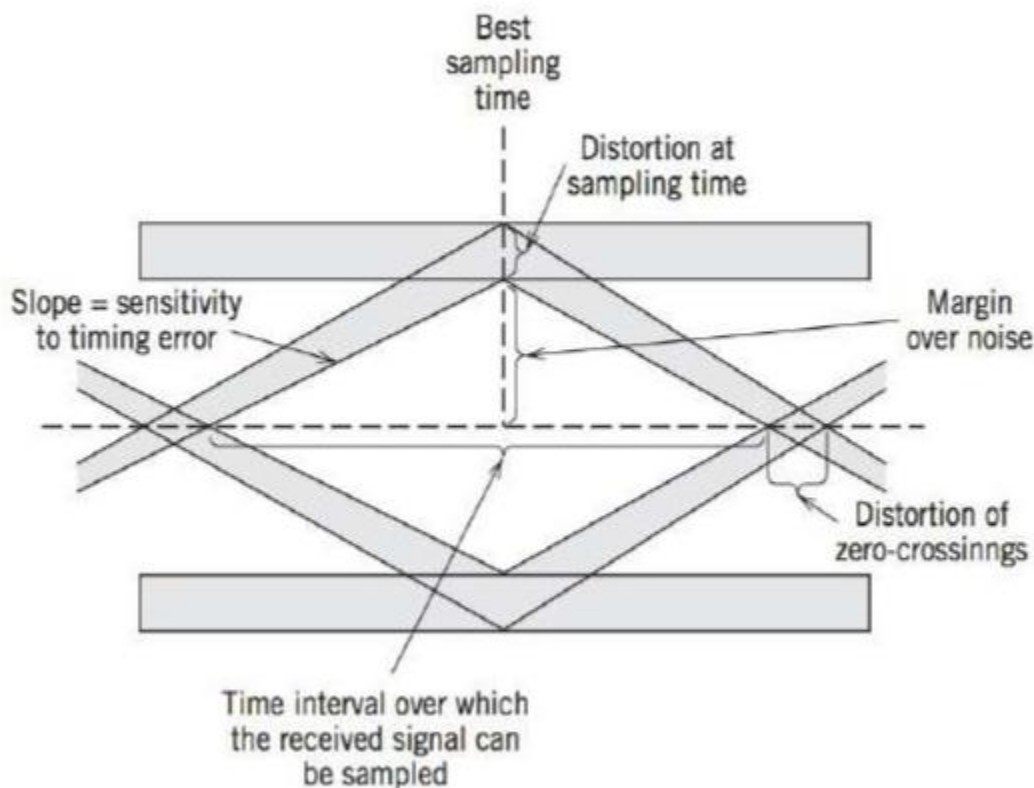
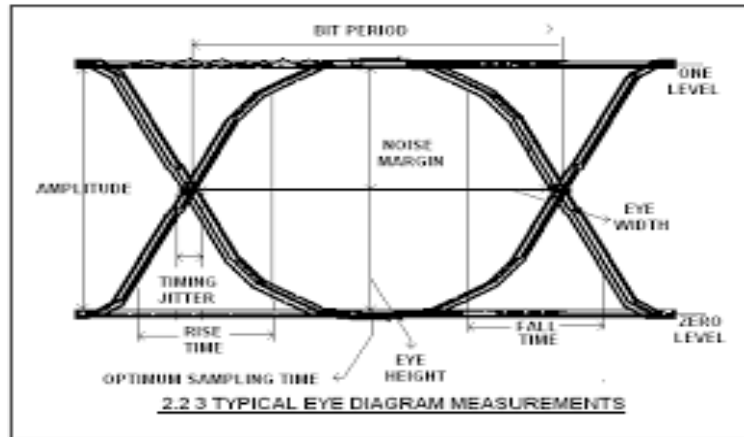
Interference Detection: It helps in identifying issues like inter symbol interference (ISI), noise, and jitter which can distort the signal.

Timing Errors: The eye diagram allows engineers to see if the timing of the signal transitions (from 0 to 1 and 1 to 0) is correct. Any deviations can indicate timing errors.

Amplitude Variations: It shows variations in the amplitude of the signal, helping to identify issues with signal strength and voltage levels.

Detection of Noise and Jitter

- **Noise:** Noise appears as fuzziness or blurring in the eye diagram. A clean eye diagram with sharp edges indicates low noise levels.
- **Jitter:** Jitter, or timing variations, can be seen in the width of the eye opening. The more consistent the signal timing, the clearer and more open the eye.
- **Eye Opening:** The width and height of the eye opening are critical metrics. A wider eye opening means the signal can tolerate more noise and jitter, indicating a more robust communication system.
- **Signal Integrity:** The size of the eye opening is directly related to the integrity and reliability of the transmitted signal.



An operational tool for evaluating the effects of ISI in an insightful manner is the so-called eye pattern. The eye pattern is defined as the synchronized superposition of all possible realizations of the signal of

interest (e.g., received signal, receiver output) viewed within a particular signaling interval. The eye pattern derives its name from the fact that it resembles the human eye for binary waves. The interior region of the eye pattern is called the eye opening

An eye pattern provides a great deal of useful information about the performance of a data transmission system, as described in Figure 8.12

- The width of the eye opening defines the time interval over which the received signal can be sampled without error from inter symbol interference. It is apparent that the preferred time for sampling is the instant of time at which the eye is open the widest.
- The sensitivity of the system to timing errors is determined by the rate of closure of the eye as the sampling time is varied
- The height of the eye opening, at a specified sampling time, defines the *noise margin* of the system.

When the effect of intersymbol interference is severe, traces from the upper portion of the eye pattern cross traces from the lower portion, with the result that the eye is completely closed. In such a situation, it is impossible to avoid errors due to the presence of intersymbol interference in the system.

NYQUIST'S CRITERION FOR DISTORTIONLESS TRANSMISSION

The Nyquist criterion for distortionless transmission, also known as the Nyquist ISI (Inter-Symbol Interference) criterion, is fundamental in digital communications. It ensures that a signal can be transmitted over a channel without introducing ISI, which occurs when the signal from one symbol interferes with adjacent symbols, leading to errors in decoding.

Key Concepts of the Nyquist Criterion

1. Nyquist Rate:

- The Nyquist rate is the minimum sampling rate required to avoid aliasing and accurately reconstruct the original signal. For a baseband signal with bandwidth B , the Nyquist rate is $2B$ samples per second.

2. Nyquist Interval:

- This is the time interval T between successive samples in a signal sampled at the Nyquist rate. For a signal with bandwidth B , the Nyquist interval is $1/2B$ seconds.

3. Zero ISI Condition:

- For distortionless transmission, the condition that the sampled signal should meet is that it should have zero ISI. This means the sampled values should be zero at all points except at the sampling instants where the signal is non-zero.

Importance

The Nyquist criterion is crucial for:

- **Bandwidth Efficiency:** Ensuring that the transmission uses the available bandwidth efficiently without wasting spectrum.
- **Error Reduction:** Minimizing ISI reduces errors in decoding, leading to more reliable communication.

In practice, we typically find that the transfer function of a channel and the transmitted pulse shape are specified, and the problem is to determine the transfer functions of the transmit and receive filters so as to reconstruct the original binary data sequence $\{b_k\}$. The receiver does this by *extracting* and then *decoding* the corresponding sequence of coefficients, $\{a_k\}$, from the output $y(t)$. The *extraction* involves sampling the output $y(t)$ at time $t = iT_b$. The *decoding* requires that the weighted pulse contribution $a_k p(iT_b - kT_b)$ for $k = i$ be *free* from ISI due to the overlapping tails of all other weighted pulse contributions represented by $k \neq i$. This, in turn, requires that we *control* the overall pulse $p(t)$, as shown by

$$p(iT_b - kT_b) = \begin{cases} 1, & i = k \\ 0, & i \neq k \end{cases} \quad (8.49)$$

where $p(0) = 1$, by normalization. If $p(t)$ satisfies the condition of Eq (8.49), the receiver output $y(t_i)$ given in Eq. (8.48) simplifies to (ignoring the noise term)

$$y(t_i) = \mu a_i \quad \text{for all } i$$

which implies zero intersymbol interference. Hence, the condition of Eq. (8.49) ensures *perfect reception in the absence of noise*.

From a design point of view, it is informative to transform the condition of Eq. (8.49) into the frequency domain. Consider then the sequence of samples $\{p(nT_b)\}$, where $n = 0, \pm 1, \pm 2, \dots$. From the discussion presented in Chapter 7 on the sampling process, we recall that sampling in the time domain produces periodicity in the frequency domain. In particular, we may write

$$P_\delta(f) = R_b \sum_{n=-\infty}^{\infty} P(f - nR_b) \quad (8.50)$$

where $R_b = 1/T_b$ is the *bit rate* in bits per second (b/s); $P_\delta(f)$ is the Fourier transform of an infinite periodic sequence of delta functions of period T_b , whose areas are weighted by the respective sample values of $p(t)$. That is, $P_\delta(f)$ is given by

$$P_\delta(f) = \int_{-\infty}^{\infty} \sum_{m=-\infty}^{\infty} [p(mT_b) \delta(t - mT_b)] \exp(-j2\pi ft) dt \quad (8.51)$$

Let the integer $m = i - k$. Then, $i = k$ corresponds to $m = 0$, and likewise $i \neq k$ corresponds to $m \neq 0$. Accordingly, imposing the condition of Eq. (8.49) on the sample values of $p(t)$ in the integral of Eq. (8.51), we get

$$\begin{aligned} P_\delta(f) &= \int_{-\infty}^{\infty} p(0) \delta(t) \exp(-j2\pi ft) dt \\ &= p(0) \end{aligned} \quad (8.52)$$

where we have made use of the sifting property of the delta function. Since from Eq. (8.46) we have $p(0) = 1$, it follows from Eqs. (8.50) and (8.52) that the condition for zero intersymbol interference is satisfied if

$$\sum_{n=-\infty}^{\infty} P(f - nR_b) = T_b \quad (8.53)$$

IDEAL NYQUIST CHANNEL

The simplest way of satisfying Eq. (8.53) is to specify the frequency function $P(f)$ to be in the form of a *rectangular function*, as shown by

$$P(f) = \begin{cases} \frac{1}{2W}, & -W < f < W \\ 0, & |f| > W \end{cases} \quad (8.54)$$

$$= \frac{1}{2W} \text{rect}\left(\frac{f}{2W}\right)$$

where the overall system bandwidth W is defined by

$$W = \frac{R_b}{2} = \frac{1}{2T_b} \quad (8.55)$$

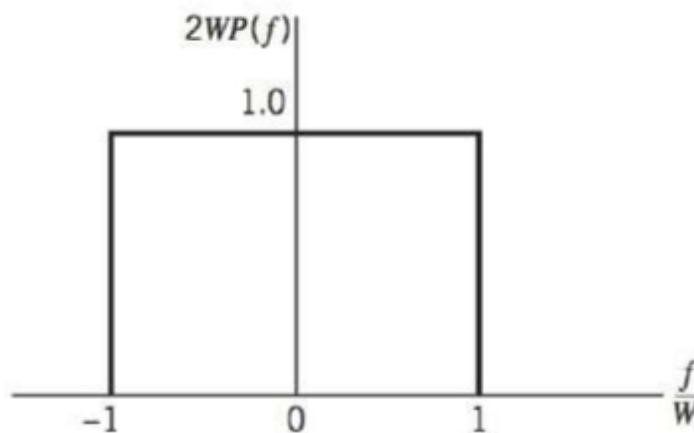
According to the solution described by Eqs. (8.54) and (8.55), no frequencies of absolute value exceeding half the bit rate are needed. Hence, one signal waveform that produces zero intersymbol interference is defined by the *sinc function*:

$$p(t) = \frac{\sin(2\pi Wt)}{2\pi Wt} \quad (8.56)$$

$$= \text{sinc}(2Wt)$$

The special value of the bit rate $R_b = 2W$ is called the *Nyquist rate*, and W is itself called the *Nyquist bandwidth*. Correspondingly, the ideal baseband pulse transmission system described by Eq. (8.54) in the frequency domain or, equivalently, Eq. (8.56) in the time domain is called the *ideal Nyquist channel*.

Figures 8.15a and 8.15b show plots of $P(f)$ and $p(t)$, respectively. In Figure 8.15a, the normalized form of the frequency function $P(f)$ is shown plotted for positive and negative frequencies. In Figure 8.15b, we have also included the signaling intervals and the corresponding centered sampling instants. The function $p(t)$ can be regarded as the impulse response of an ideal low-pass filter with passband amplitude response $1/2W$ and bandwidth W . The function $p(t)$ has its peak value at the origin and goes through zero at integer multiples of the bit duration T_b . It is apparent that if the received waveform $y(t)$ is sampled at the instants of time $t = 0, \pm T_b, \pm 2T_b, \dots$, then the pulses defined by $\mu p(t - iT_b)$ with arbitrary amplitude μ and index $i = 0, \pm 1, \pm 2, \dots$, will not interfere with each other. This condition is illustrated in Figure 8.16 for the binary sequence 1011010.



(a)

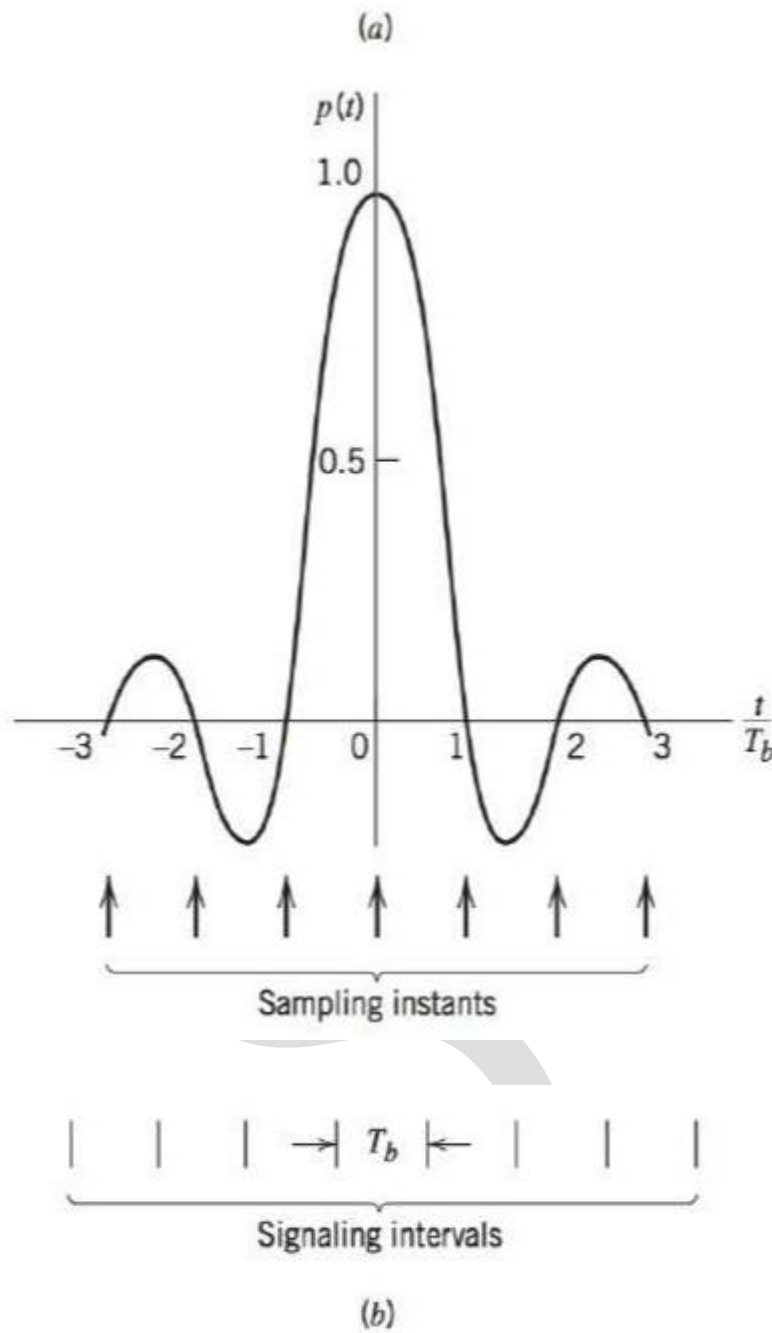


FIGURE 8.15 (a) Ideal amplitude response.
(b) Ideal basic pulse shape.

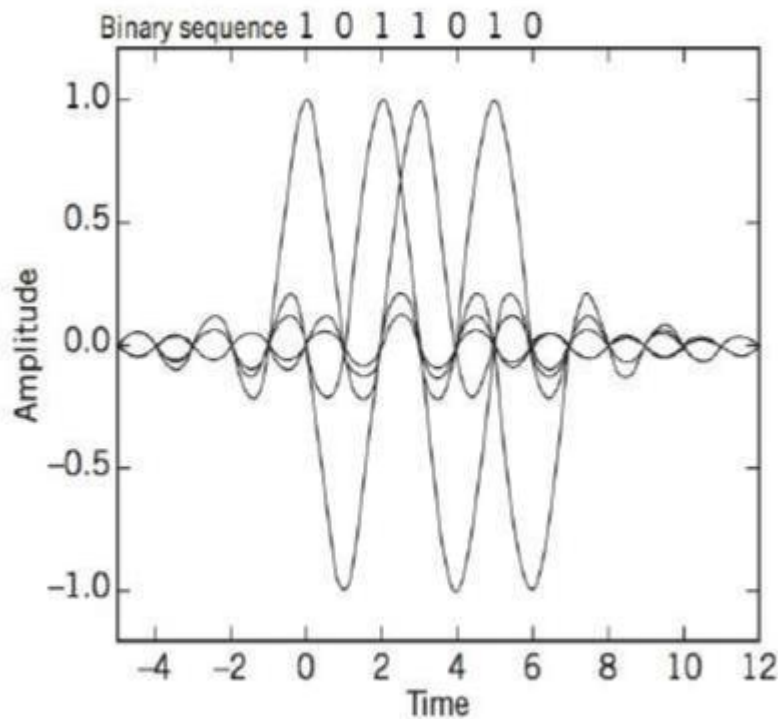


FIGURE 8.16 A series of sine pulses corresponding to the sequence 1011010.

Although the use of the ideal Nyquist channel does indeed achieve economy in bandwidth in that it solves the problem of zero intersymbol interference with the minimum bandwidth possible, there are two practical difficulties that make it an undesirable objective for system design:

1. It requires that the amplitude characteristic of $P(f)$ be flat from $-W$ to W , and zero elsewhere. This is physically unrealizable because of the abrupt transitions at the band edges $\pm W$.
2. The function $p(t)$ decreases as $1/|t|$ for large $|t|$, resulting in a slow rate of decay. This is also caused by the discontinuity of $P(f)$ at $\pm W$. Accordingly, there is practically no margin of error in sampling times in the receiver.

To evaluate the effect of this *timing error*, consider the sample of $y(t)$ at $t = \Delta t$, where Δt is the timing error. To simplify the exposition, we may put the correct sampling time t_i equal to zero. In the absence of noise, we thus have

$$\begin{aligned} y(\Delta t) &= \mu \sum_k a_k p(\Delta t - kT_b) \\ &= \mu \sum_k a_k \frac{\sin[2\pi W(\Delta t - kT_b)]}{2\pi W(\Delta t - kT_b)} \end{aligned} \quad (8.57)$$

Since $2WT_b = 1$, by definition, we may rewrite Eq. (8.57) as

$$y(\Delta t) = \mu a_0 \operatorname{sinc}(2\pi W \Delta t) + \frac{\mu \sin(2\pi W \Delta t)}{\pi} \sum_{\substack{k \\ k \neq 0}} \frac{(-1)^k a_k}{(2W \Delta t - k)} \quad (8.58)$$

The first term on the right-hand side of Eq. (8.58) defines the desired symbol, whereas the remaining series represents the intersymbol interference caused

by the timing error Δt in sampling the output $y(t)$. Unfortunately, it is possible for this series to diverge, thereby causing erroneous decisions in the receiver.

RAISED COSINE SPECTRUM

We may overcome the practical difficulties encountered with the ideal Nyquist channel by extending the bandwidth from the minimum value $W = R_b/2$ to an adjustable value between W and $2W$. We now specify the frequency function $P(f)$ to satisfy a condition more elaborate than that for the ideal Nyquist channel. Specifically, we retain three terms of Eq. (8.53) and restrict the frequency band of interest to $[-W, W]$, as shown by

$$P(f) + P(f - 2W) + P(f + 2W) = \frac{1}{2W}, \quad -W \leq f \leq W \quad (8.59)$$

We may devise several band-limited functions that satisfy Eq. (8.59). A particular form of $P(f)$ that embodies many desirable features is provided by a *raised cosine spectrum*. This frequency characteristic consists of a flat portion and a *rolloff* portion that has a sinusoidal form, as follows:

$$P(f) = \begin{cases} \frac{1}{2W}, & 0 \leq |f| < f_1 \\ \frac{1}{4W} \left\{ 1 - \sin \left[\frac{\pi(|f| - W)}{2W - 2f_1} \right] \right\}, & f_1 \leq |f| < 2W - f_1 \\ 0, & |f| \geq 2W - f_1 \end{cases} \quad (8.60)$$

The frequency parameter f_1 and bandwidth W are related by

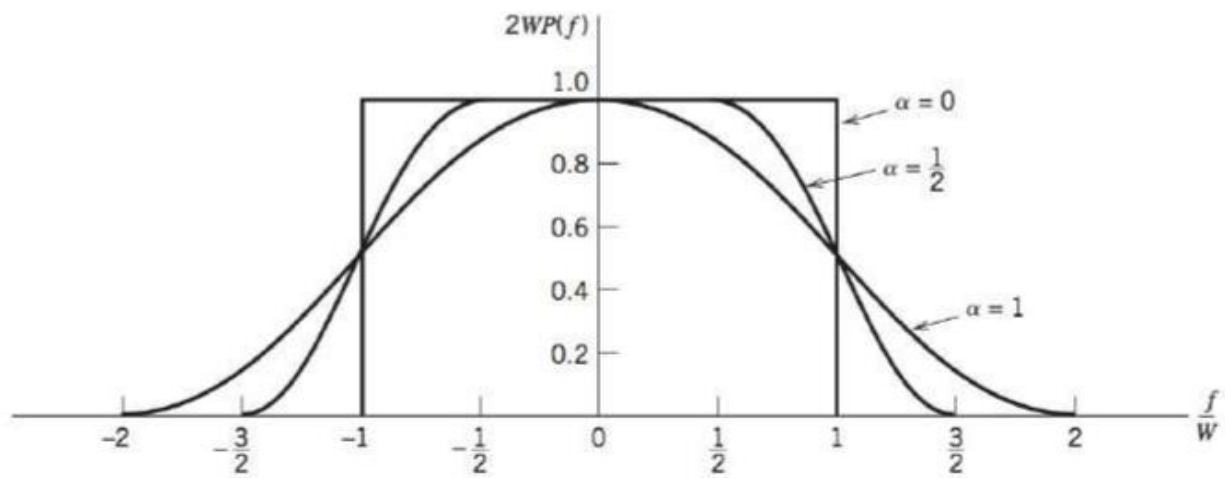
$$\alpha = 1 - \frac{f_1}{W} \quad (8.61)$$

The parameter α is called the *rolloff factor*; it indicates the *excess bandwidth* over the ideal solution, W . Specifically, the transmission bandwidth B_T is defined by $2W - f_1 = W(1 + \alpha)$.

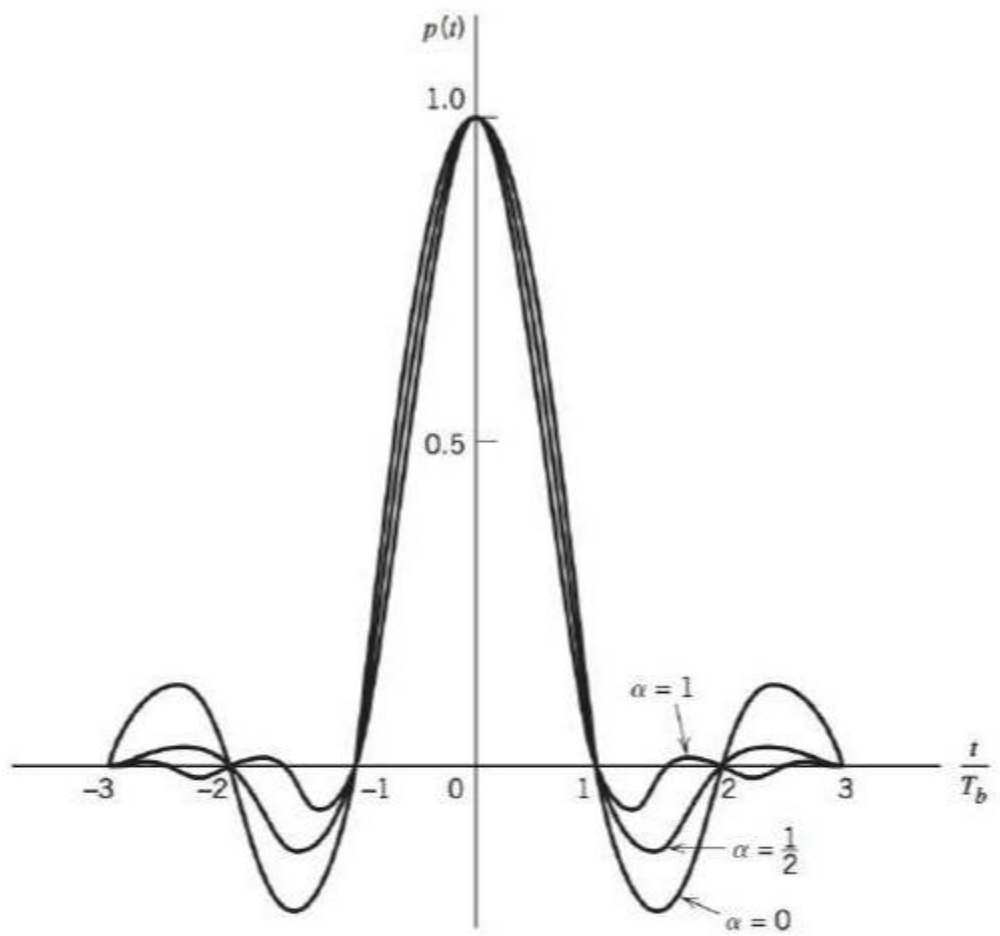
The frequency response $P(f)$, normalized by multiplying it by $2W$, is shown plotted in Figure 8.17a for three values of α , namely, 0, 0.5, and 1. We see that for $\alpha = 0.5$ or 1, the function $P(f)$ cuts off gradually as compared with the ideal Nyquist channel (i.e., $\alpha = 0$) and is therefore easier to implement in practice. Also the function $P(f)$ exhibits odd symmetry with respect to the Nyquist bandwidth W , making it possible to satisfy the condition of Eq. (8.59).

The time response $p(t)$ is the inverse Fourier transform of the function $P(f)$. Hence, using the $P(f)$ defined in Eq. (8.60), we obtain the result (see Problem 8.10)

$$p(t) = \left[\text{sinc}(2Wt) \right] \left(\frac{\cos(2\pi\alpha Wt)}{1 - 16\alpha^2 W^2 t^2} \right) \quad (8.62)$$



(a)



The special case with $\alpha = 1$ (i.e., $f_1 = 0$) is known as the *full-cosine rolloff* characteristic, for which the frequency response of Eq. (8.60) simplifies to

$$P(f) = \begin{cases} \frac{1}{4W} \left[1 + \cos\left(\frac{\pi f}{2W}\right) \right], & 0 < |f| < 2W \\ 0, & |f| \geq 2W \end{cases} \quad (8.63)$$

Correspondingly, the time response $p(t)$ simplifies to

$$p(t) = \frac{\text{sinc}(4Wt)}{1 - 16W^2t^2} \quad (8.64)$$

This time response exhibits two interesting properties:

1. At $t = \pm T_b/2 = \pm 1/4W$, we have $p(t) = 0.5$; that is, the pulse width measured at half amplitude is exactly equal to the bit duration T_b .
2. There are zero crossings at $t = \pm 3T_b/2, \pm 5T_b/2, \dots$ in addition to the usual zero crossings at the sampling times $t = \pm T_b, \pm 2T_b, \dots$

These two properties are extremely useful in extracting a timing signal from the received

BASEBAND M-ARY PAM TRANSMISSION

In the baseband binary PAM system, the pulse-amplitude modulator produces binary pulses, that is, pulses with one of two possible amplitude levels. On the other hand, in a baseband M-ary PAM system, the pulse-amplitude modulator produces one of M possible amplitude levels, with $M > 2$, as illustrated in Figure below shows the case of a quaternary $\{M = 4\}$ system and the binary data sequence 0010110111. The corresponding electrical representation for each of the four possible pairs of bits is shown in (b).

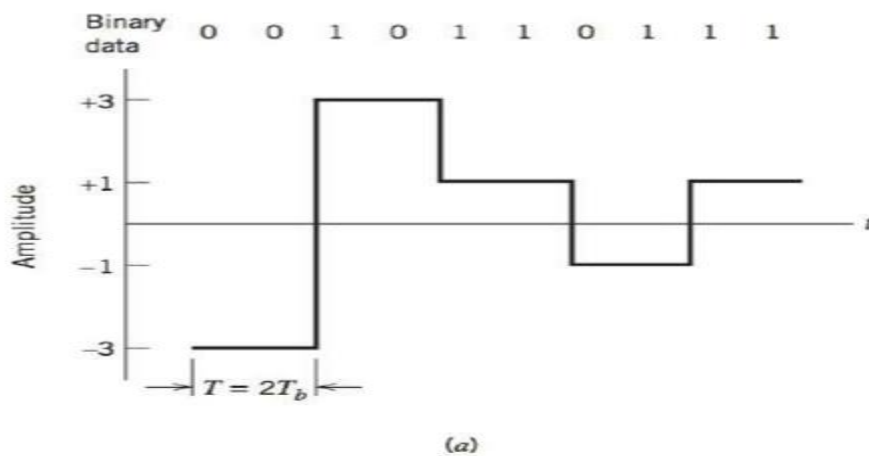
In an M-ary system, the information source emits a sequence of symbols from an alphabet that consists of M symbols. Each amplitude level at the pulse-amplitude modulator output corresponds to a distinct symbol, so that there are M distinct amplitude levels to be transmitted.

Consider an M-ary PAM system with a signal alphabet that contains M equally likely and statistically independent symbols, with the symbol duration denoted by T seconds. We refer to $1/T$ as the signaling rate of the system, which is expressed in symbols per second or bauds. It is informative to relate the signaling rate of this system to that of an equivalent binary PAM system for which the value of M is 2 and the successive binary symbols 1 and 0 are equally likely and

statistically independent with the duration of either symbol denoted by T_b seconds. Under the conditions described here, the binary PAM system produces information at the rate of $1/T_b$ bits per second. We also observe that in the case of a quaternary PAM system, for example, the four possible symbols may be identified with the bit pairs 00, 01, 10, and 11. We thus see that each symbol represents 2 bits of information, and 1 baud is equal to 2 bits per second. We may generalize this result by stating that in an M -ary PAM system, one baud is equal to $\log_2 M$ bits per second, and the symbol duration T of the M -ary PAM system is related to the bit duration T_b of the equivalent binary PAM system as

$$T = T_b \log_2 M$$

Therefore, in a given channel bandwidth, we find that by using an M -ary PAM system, we are able to transmit information at a rate that is $\log_2 M$ faster than the corresponding binary PAM system.



Dibit	Amplitude
00	-3
01	-1
11	+1
10	+3

(b)

Noise

Noise is an electronic signal that is a mixture of many random frequencies many amplitudes that gets added to a radio or information signal as it is transmitted from one place to another or as it is processed. Noise is *not* the same as interference from other information signals.

When you turn on any AM, FM, or shortwave receiver and tune it to some position between stations, the hiss or static that you hear in the speaker is noise. Noise also shows up on a black-and-white TV screen as snow or on a color screen as confetti. If the noise level is high enough and/or the signal is weak enough, the noise can completely obliterate the original signal. Noise that occurs in transmitting digital data causes *bit errors* and can result in information being garbled or lost.

The noise level in a system is proportional to temperature and bandwidth, and to the amount of current flowing in a component, the gain of the circuit, and the resistance of the circuit. Increasing any of these factors increases noise. Therefore, low noise is best obtained by using low-gain circuits, low direct current, low resistance values, and narrow bandwidths. Keeping the temperature low can also help.

Noise is a problem in communication systems whenever the received signals are very low in amplitude. When the transmission is over short distances or high-power transmitters are being used, noise is not usually a problem. But in most communication systems, weak signals are normal, and noise must be taken into account at the design stage. It is in the receiver that noise is the most detrimental because the receiver must amplify the weak signal and recover the information reliably.

Noise can be external to the receiver or originate within the receiver itself. Both types are found in all receivers, and both affect the SNR.

The *signal-to-noise (S/N) ratio*, also designated SNR, indicates the relative strengths of the signal and the noise in a communication system. The stronger the signal and the weaker the noise, the higher the *S/N* ratio. If the signal is weak and the noise is strong, the *S/N* ratio will be low and reception will be unreliable.

Communication equipment is designed to produce the highest feasible S/N ratio.

Signals can be expressed in terms of voltage or power. The S/N ratio is computed by using either voltage or power values:

$$\frac{S}{N} = \frac{V_s}{V_n} \quad \text{or} \quad \frac{S}{N} = \frac{P_s}{P_n}$$

where V_s = signal voltage
 V_n = noise voltage
 P_s = signal power
 P_n = noise power

Assume, e.g., that the signal voltage is $1.2 \mu\text{V}$ and the noise is $0.3 \mu\text{V}$. The S/N ratio is $1.2/0.3 = 4$

$$\text{For voltage: dB} = 20 \log \frac{S}{N} = 20 \log 4 = 20(0.602) = 12 \text{ dB}$$

External Noise

External noise comes from sources over which we have little or no control— industrial, atmospheric, or space. Regardless of its source, noise shows up as a random ac voltage and can be seen on an oscilloscope. The amplitude varies over a wide range, as does the frequency. Noise in general contains all frequencies, varying randomly. This is generally known as **white noise**.

Atmospheric noise and space noise are a fact of life and simply cannot be eliminated. Some industrial noise can be controlled at the source, but because there are so many sources of this type of noise, there is no way to eliminate it. The key to reliable communication, then, is simply to generate signals at a high enough power to overcome external noise. In some cases, shielding sensitive circuits in metallic enclosures can aid in noise control.

Industrial Noise. *Industrial noise* is produced by manufactured equipment, such as automotive ignition systems, electric motors, and generators. Any electrical equipment that causes high voltages or currents to be switched produces transients that create noise. Noise pulses of large

amplitude occur whenever a motor or other inductive device is turned on or off. The resulting transients are extremely large in amplitude and rich in random harmonics. Fluorescent and other forms of gas-filled lights are another common source of industrial noise.

Atmospheric Noise. The electrical disturbances that occur naturally in the earth's atmosphere are another source of noise. *Atmospheric noise* is often referred to as *static*. Static usually comes from lightning, the electric discharges that occur between clouds or between the earth and clouds. Huge static charges build up on the clouds, and when the potential difference is great enough, an arc is created and electricity literally flows through the air. Lightning is very much like the static charges that we experience during a dry spell in the winter. The voltages involved are, however, enormous, and these transient electric signals of megawatt power generate harmonic energy that can travel over extremely long distances.

Like industrial noise, atmospheric noise shows up primarily as amplitude variations that add to a signal and interfere with it. Atmospheric noise has its greatest impact on signals at frequencies below 30 MHz.

Extraterrestrial Noise. *Extraterrestrial noise*, solar and cosmic, comes from sources in space. One of the primary sources of extraterrestrial noise is the sun, which radiates a wide range of signals in a broad noise spectrum. The noise intensity produced by the sun varies with time. In fact, the sun has a repeatable 11-year noise cycle. During the peak of the cycle, the sun produces an awesome amount of noise that causes tremendous radio signal interference and makes many frequencies unusable for communication. During other years, the noise is at a lower level.

Noise generated by stars outside our solar system is generally known as *cosmic noise*. Although its level is not as great as that of noise produced by the sun, because of the great distances between those stars and earth, it is nevertheless an important source of noise that must be considered. It shows up primarily in the 10-MHz to 1.5-GHz range, but causes the greatest disruptions in the 15- to 150-MHz range.

Internal Noise

Electronic components in a receiver such as resistors, diodes, and transistors are major sources of *internal noise*. Internal noise, although it is low level, is often great enough to interfere with weak signals. The main sources of internal noise in a receiver are thermal noise, semiconductor noise, and intermodulation distortion. Since the sources of internal noise are well known, there

is some design control over this type of noise.

Thermal Noise. Most internal noise is caused by a phenomenon known as *thermal agitation*, the random motion of free electrons in a conductor caused by heat. Increasing the temperature causes this atomic motion to increase. Since the components are conductors, the movement of electrons constitutes a current flow that causes a small voltage to be produced across that component. Electrons traversing a conductor as current flows experience fleeting impediments in their path as they encounter the thermally agitated atoms. The apparent resistance of the conductor thus fluctuates, causing the thermally produced random voltage we call noise.

This noise can be noticed by simply connecting a high-value (megohm) resistor to a very high-gain oscilloscope. The motion of the electrons due to room temperature in the resistor causes a voltage to appear across it. The voltage variation is completely random and at a very low level. The noise developed across a resistor is proportional to the temperature to which it is exposed.

Thermal agitation is often referred to as *white noise* or *Johnson noise*, after J. B. Johnson, who discovered it in 1928. Just as white light contains all other light frequencies, white noise contains all frequencies randomly occurring at random amplitudes. A white noise signal therefore occupies, theoretically at least, infinite bandwidth. Filtered or band-limited noise is referred to as *pink noise*.

In a relatively large resistor at room temperature or higher, the noise voltage across it can be as high as several microvolts. This is the same order of magnitude as or higher than that of many weak RF signals. Weaker-amplitude signals will be totally masked by this noise.

Since noise is a very broadband signal containing a tremendous range of random frequencies, its level can be reduced by limiting the bandwidth. If a noise signal is fed into a selective tuned circuit, many of the noise frequencies are rejected and the overall noise level goes down. The noise power is proportional to the bandwidth of any circuit to which it is applied. Filtering can reduce the noise level, but does not eliminate it entirely.

The amount of open-circuit noise voltage appearing across a resistor or the input impedance to a receiver can be calculated according to Johnson's formula

$$v_n = \sqrt{4kTBR}$$

where v_n = rms noise voltage

k = Boltzman's constant (1.38×10^{-23} J/K)

T = temperature, K ($^{\circ}\text{C} + 273$)

B = bandwidth, Hz

R = resistance, Ω

Example

What is the open-circuit noise voltage across a $100\text{-k}\Omega$ resistor over the frequency range of direct current to 20 kHz at room temperature (25°C)?

$$\begin{aligned} v_n &= \sqrt{4kTBR} \\ &= \sqrt{4(1.38 \times 10^{-23})(25 + 273)(20 \times 10^3)(100 \times 10^3)} \\ v_n &= 5.74 \mu\text{V} \end{aligned}$$

Example

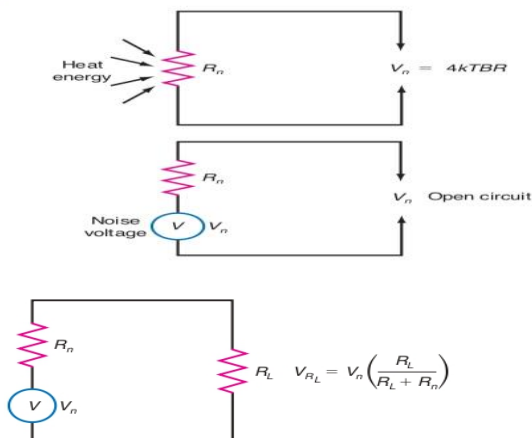
The bandwidth of a receiver with a $75\text{-}\Omega$ input resistance is 6 MHz. The temperature is 29°C . What is the input thermal noise voltage?

$$T = 29 + 273 = 302 \text{ K}$$

$$v_n = \sqrt{4kTBR}$$

$$v_n = \sqrt{4(1.38 \times 10^{-23})(302)(6 \times 10^6)(75)} = 2.74 \mu\text{V}$$

Figure 9-22 A resistor acts as a tiny generator of noise voltage.



TEMPERATURE SCALES AND CONVERSIONS

When calculating noise values, you will frequently need to make conversions from one of these temperature scales to another. The most common conversion formulas are given here.

$$T_C = 5(T_F - 32)/9 \quad \text{or} \quad T_C = T_K - 273$$

$$T_F = \frac{9T_C}{5} + 32$$

$$T_K = T_C + 273$$

Since noise voltage is proportional to resistance value, temperature, and bandwidth, noise voltage can be reduced by reducing resistance, temperature, and bandwidth or any combination to the minimum level acceptable for the given application. In many cases, of course, the values of resistance and bandwidth cannot be changed. One thing, however, that is always controllable to some extent is temperature. Anything that can be done to cool the circuits will greatly reduce the noise. Heat sinks, cooling fans, and good ventilation can help lower noise. Many low-noise receivers for weak microwave signals from spacecraft and in radio telescopes are supercooled; i.e., their temperature is reduced to very low (cryogenic) levels with liquid nitrogen or liquid helium.

Thermal noise can also be computed as a power level. Johnson's formula is then

$$P_n = kTB$$

where P_n is the average noise power in watts.

Note that when you are dealing with power, the value of resistance does not enter into the equation.

Example

What is the average noise power of a device operating at a temperature of 90°F with a bandwidth of 30 kHz?

$$T_C = 5(T_F - 32)/9 = 5(90 - 32)/9 = 5(58)/9 = 290/9 = 32.2^\circ\text{C}$$

$$T_K = T_C + 273 = 32.2 + 273 = 305.2 \text{ K}$$

$$P_n = (1.38 \times 10^{-23})(305.2)(30 \times 10^3) = 1.26 \times 10^{-16} \text{ W}$$

Semiconductor Noise. Electronic components such as diodes and transistors are major contributors of noise. In addition to thermal noise, semiconductors produce shot noise, transit-time noise, and flicker noise.

The most common type of *semiconductor noise* is *shot noise*. Current flow in any device is not direct and linear. The current carriers, electrons or holes, sometimes take random paths from source to destination, whether the destination is an output element, tube plate, or collector or drain in a transistor. It is this random movement that produces the shot effect. Shot noise is also produced by the random movement of electrons or holes across a PN junction. Even though current flow is established by external bias voltages, some random movement of electrons or holes will occur as a result of discontinuities in the device. For example, the interface between the copper lead and the semiconductor material forms a discontinuity that causes random movement of the current carriers.

Shot noise is also white noise in that it contains all frequencies and amplitudes over a very wide range. The amplitude of the noise voltage is unpredictable, but it does follow a Gaussian distribution curve that is a plot of the probability that specific amplitudes will occur. The amount of shot noise is directly proportional to the amount of dc bias flowing in a device. The bandwidth of the device or circuit is also important. The rms noise current in a device I_n is calculated with the formula

$$I_n = \sqrt{2qIB}$$

where q = charge on an electron, 1.6×10^{-19} C

I = direct current, A

B = bandwidth, Hz

As an example, assume a dc bias of 0.1 mA and a bandwidth of 12.5 kHz. The noise current is

$$I_n = \sqrt{2(1.6 \times 10^{-19})(0.0001)(12,500)} = \sqrt{4 \times 10^{-19}} = 0.632 \times 10^{-9} \text{ A}$$

$$I_n = 0.632 \text{ nA}$$

Now assume that the current is flowing across the emitter-base junction of a bipolar transistor. The dynamic resistance of this junction r_e' can be calculated with the expression $r_e' = 0.025/I_e$, where I_e is the emitter current. Assuming an emitter current of 1 mA, we have $r_e' = 0.025/0.001 = 25 \Omega$. The noise voltage across the junction is found with Ohm's law:

$$v_n = I_n r_e' = 0.623 \times 10^{-9} \times 25 = 15.8 \times 10^{-9} \text{ V} = 15.8 \text{ nV}$$

Another kind of noise that occurs in transistors is called *transit-time noise*. The term *transit time* refers to how long it takes for a current carrier such as a hole or electron to move from the input to the output. The devices themselves are very tiny, so the distances involved are minimal, yet the time it takes for the current carriers to move even a short distance is finite. At low frequencies, this time is negligible; but when the frequency of operation is high and the period of the signal being processed is the same order of magnitude as the transit time, problems can occur. Transit-time noise shows up as a kind of random variation of current carriers within a device, occurring near the upper cutoff frequency. Transit-time noise is directly proportional to the frequency of operation. Since most circuits are designed to operate at a frequency much less than the transistor's upper limit, transit-time noise is rarely a problem.

A third type of semiconductor noise, *flicker noise* or excess noise, also occurs in resistors and conductors. This disturbance is the result of minute random variations of resistance in the semiconductor material. It is directly proportional to current and temperature. However, it is inversely proportional to frequency, and for this reason it is sometimes referred to as $1/f$ noise. Flicker noise is highest at the lower frequencies and thus is not pure white noise. Because of the dearth of high-frequency components, $1/f$ noise is also called *pink noise*.

Figure 9-24 Noise in a transistor with respect to frequency.

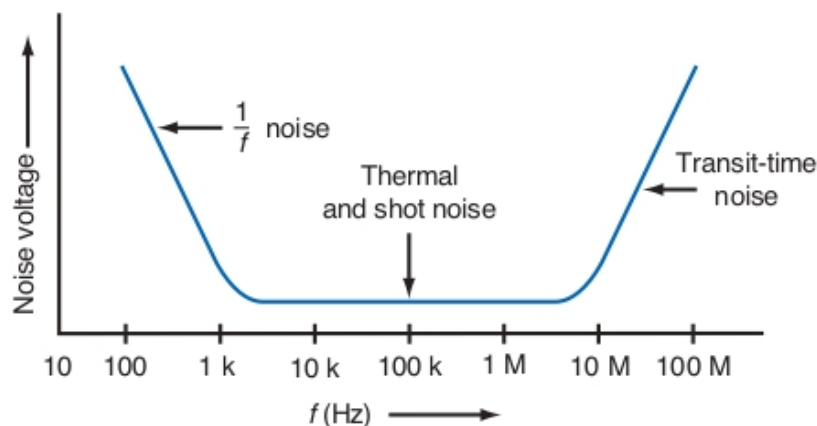


Fig. 9-24 shows the total noise voltage variation in a transistor, which is a composite of the various noise sources. At low frequencies, noise voltage is high, because of $1/f$ noise. At very high frequencies, the rise in noise is due to transit-time effects near the upper cutoff frequency of the device. Noise is lowest in the midrange, where most devices operate. The noise in this range is due to thermal and shot effects, with shot noise sometimes contributing more than thermal noise.

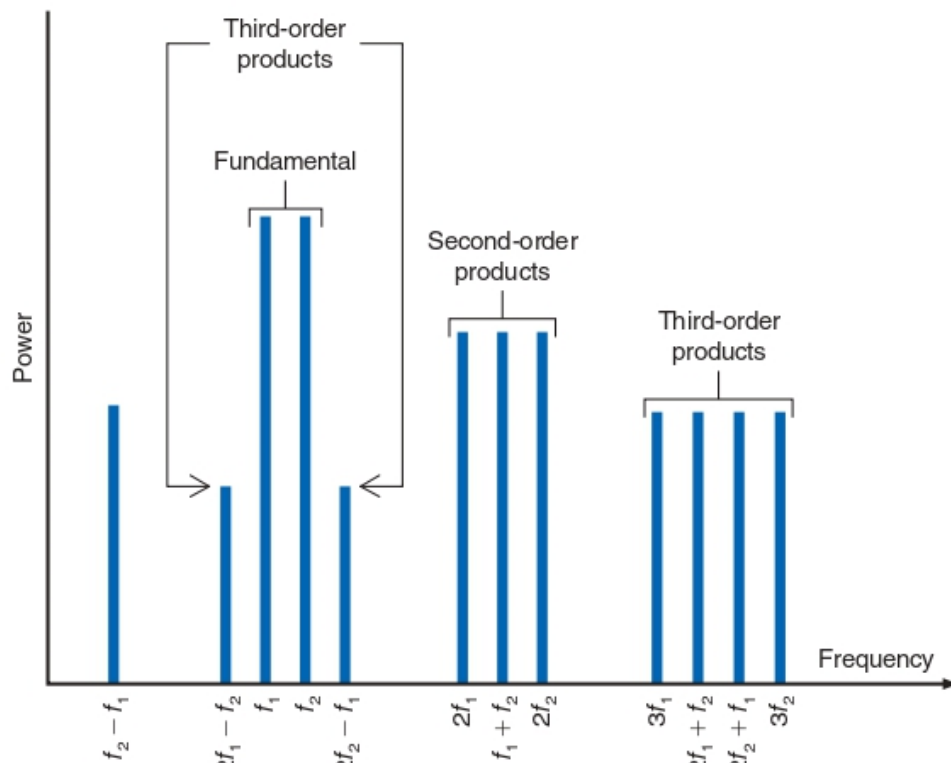
Intermodulation Distortion. *Intermodulation distortion* results from the generation of new signals and harmonics caused by circuit nonlinearities. As stated previously, circuits can never be perfectly linear, and if bias voltages are incorrect in an amplifier or it is driven into clipping, it is likely to be more nonlinear than intended.

Nonlinearities produce modulation or heterodyne effects. Any frequencies in the circuit mix together, forming sum and difference frequencies. When many frequencies are involved, or with pulses or rectangular waves, the large number of harmonics produces an even larger number of sum and difference frequencies.

When two signals are near the same frequency, some new sum and difference frequencies are generated by a nonlinearity, and they can appear inside the bandwidth of the amplifier. In most cases, such signals cannot be filtered out. As a result they become interfering signals to the primary signals to be amplified. They are a form of noise.

Fig. 9-25 illustrates this. Signals f_1 and f_2 appear within the bandwidth of an amplifier. Any nonlinearity generates new signals $f_1 - f_2$ and $f_1 + f_2$. In addition, these new signals

Figure 9-25 An illustration of intermodulation distortion products produced from two input signals f_1 and f_2 amplifier nonlinearities.



begin to mix with some of the harmonics generated by the nonlinearity ($2f_1$, $2f_2$, $3f_1$, $3f_2$, etc.). Some of these new signals will occur within the amplifier band pass. Those new signals that cause the most trouble are the so-called third-order products, specifically $2f_1 \pm f_2$ and $2f_2 \pm f_1$. See the figure. Those most likely to be in the amplifier bandwidth are $2f_1 - f_2$ and $2f_2 - f_1$. These are the third-order products. The key to minimizing these extraneous intermodulation products is to maintain good linearity through biasing and input signal level control.

Expressing Noise Levels

The noise quality of a receiver can be expressed as in terms of noise figure, noise factor, noise temperature, and SINAD.

Noise Factor and Noise Figure. The *noise factor* is the ratio of the S/N power at the input to the S/N power at the output. The device under consideration can be the entire receiver or a single amplifier stage. The noise factor or noise ratio (NR) is computed with the expression

$$\text{NR} = \frac{S/N \text{ input}}{S/N \text{ output}}$$

When the noise factor is expressed in decibels, it is called the *noise figure (NF)*:

$$\text{NF} = 10 \log \text{NR} \quad \text{dB}$$

Amplifiers and receivers always have more noise at the output than at the input because of the internal noise, which is added to the signal. And even as the signal is being amplified along the way, the noise generated in the process is amplified along with it. The S/N ratio at the output will be less than the S/N ratio of the input, and so the noise figure will always be greater than 1. A receiver that contributed no noise to the signal would have a noise figure of 1, or 0 dB, which is not attainable in practice. A transistor amplifier in a communication receiver usually has a noise figure of several decibels. The lower the noise figure, the better the amplifier or receiver. Noise figures of less than about 2 dB are excellent.

Example

An RF amplifier has an S/N ratio of 8 at the input and an S/N ratio of 6 at the output. What are the noise factor and noise figure?

$$\text{NR} = \frac{8}{6} = 1.333$$

$$\text{NF} = 10 \log 1.3333 = 10(0.125) = 1.25 \text{ dB}$$

Noise Temperature. Most of the noise produced in a device is thermal noise, which is directly proportional to temperature. Therefore, another way to express the noise in an amplifier or receiver is in terms of *noise temperature* T_N . Noise temperature is expressed in kelvins. Remember that the Kelvin temperature scale is related to the Celsius scale by the relationship $T_K = T_C + 273$. The relationship between noise temperature and NR is given by

$$T_N = 290(\text{NR} - 1)$$



Example

A receiver with a $75\text{-}\Omega$ input resistance operates at a temperature of 31°C . The received signal is at 89 MHz with a bandwidth of 6 MHz . The received signal voltage of $8.3\text{ }\mu\text{V}$ is applied to an amplifier with a noise figure of 2.8 dB . Find (a) the input noise power, (b) the input signal power, (c) S/N , in decibels, (d) the noise factor and S/N of the amplifier, and (e) the noise temperature of the amplifier.

a. $T_C = 273 + 31 = 304\text{ K}$

$$v_n = \sqrt{4kTBR}$$

$$v_n = \sqrt{4(1.38 \times 10^{-23})(304)(6 \times 10^6)(75)} = 2.75\text{ }\mu\text{V}$$

$$P_n = \frac{(v_n)^2}{R} = \frac{(2.75 \times 10^{-6})^2}{75} = 0.1\text{ pW}$$

b. $P_s = \frac{(v_s)^2}{R} = \frac{(8.3 \times 10^{-6})^2}{75} = 0.918\text{ pW}$

c. $\frac{S}{N} = \frac{P_s}{P_n} = \frac{0.918}{0.1} = 9.18$

$$\text{dB} = 10 \log \frac{S}{N} = 10 \log 9.18$$

$$\frac{S}{N} = 9.63\text{ dB}$$

d. $\text{NF} = 10 \log \text{NR}$

$$\text{NR} = \text{antilog} \frac{\text{NF}}{10} = 10^{\text{NF}/10}$$

$$\text{NF} = 2.8\text{ dB}$$

$$\text{NR} = 10^{2.8/10} = 10^{0.28} = 1.9$$

$$\text{NR} = \frac{S/N \text{ input}}{S/N \text{ output}}$$

$$\frac{S}{N} \text{ output} = \frac{S/N \text{ input}}{\text{NR}} = \frac{9.18}{1.9} = 4.83$$

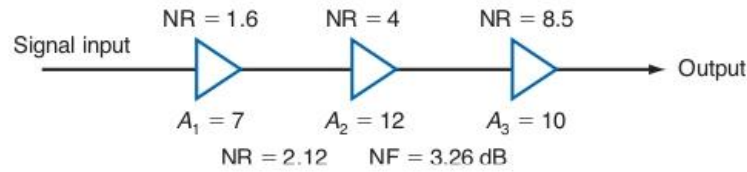
$$\frac{S}{N} (\text{output}) = \frac{S}{N} (\text{amplifier})$$

e. $T_N = 290(\text{NR} - 1) = 290(1.9 - 1) = 290(0.9) = 261\text{ K}$

Noise in Cascaded Stages

Noise has its greatest effect at the input to a receiver simply because that is the point at which the signal level is lowest. The noise performance of a receiver is invariably determined in the very first stage of the receiver, usually an RF amplifier or mixer. Design of these circuits must ensure the use of very low-noise components, taking into consideration current, resistance, bandwidth, and gain figures in the circuit. Beyond the first and second stages, noise is basically no longer a problem.



Figure 9-26 Noise in cascaded stages of amplification.

The formula used to calculate the overall noise performance of a receiver or of multiple stages of RF amplification, called *Friis' formula*, is

$$NR = NR_1 + \frac{NR_2 - 1}{A_1} + \frac{NR_3 - 1}{A_1 A_2} + \cdots + \frac{NR_n - 1}{A_1 A_2 \cdots A_{n-1}}$$

where NR = noise ratio

NR_1 = noise ratio of input or first amplifier to receive the signal

NR_2 = noise ratio of second amplifier

NR_3 = noise ratio of third amplifier, and so on

A_1 = power gain of first amplifier

A_2 = power gain of second amplifier

A_3 = power gain of third amplifier, and so on

Note that the noise ratio is used, rather than the noise figure, and so the gains are given in power ratios rather than in decibels.

As an example, consider the circuit shown in Fig. 9-26. The overall noise ratio for the combination is calculated as follows:

$$NR = 1.6 + \frac{4 - 1}{7} + \frac{8.5 - 1}{(7)(12)} = 1.6 + 0.4286 + 0.0893 = 2.12$$

The noise figure is

$$NF = 10 \log NR = 10 \log 2.12 = 10(0.326) = 3.26 \text{ dB}$$

What this calculation means is that *the first stage controls the noise performance for the whole amplifier chain*. This is true even though stage 1 has the lowest NR, because after the first stage, the signal is large enough to overpower the noise. This result is true for almost all receivers and other equipment incorporating multistage amplifiers.